

Problems in Thermodynamics 2

- 1) Problem 20 in the textbook volume 2 chapter 2.
- 2) Problem 22 in the textbook volume 2 chapter 2.
- 3) Problem 30 in the textbook volume 2 chapter 2.
- 4) Problem 39 in the textbook volume 2 chapter 2.
- 5) Problem 42 in the textbook volume 2 chapter 2.
- 6) Problem 58 in the textbook volume 2 chapter 2.
- 7) Problem 70 in the textbook volume 2 chapter 2.

You may use the trapezoid method in `scipy.integrate`.

Call it with two vectors `y` and `x` where `y` is the dependent variable and `x` the independent variable (the order is important). It returns an approximation to the integral $\int y dx$.